

The Solar System

Our Place in Space

The solar system is made up of the Sun and everything that travels around it. This includes eight planets, their moons, dwarf planets, asteroids, and comets. The Sun sits at the center of the solar system. It is a star — a giant ball of hot, glowing gas — and it contains more than 99 percent of all the mass in the solar system.

Gravity: The Force That Holds It All Together

Why don't the planets drift away into space? The answer is gravity. Gravity is a pulling force between objects. The Sun is so massive that its gravitational pull keeps all eight planets moving in orbit around it. An orbit is the curved path an object follows as it travels around another object. Without the Sun's gravity, the planets would fly off in straight lines into deep space. Gravity also keeps moons orbiting their planets, just as Earth's gravity keeps the Moon circling us.

The Eight Planets in Order

The eight planets, in order from closest to the Sun to farthest away, are: Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. A helpful way to remember the order is the sentence: 'My Very Excellent Mother Just Served Us Noodles.' Mercury is the smallest planet and the closest to the Sun. Neptune is the farthest planet, so far away that it takes 165 Earth years to complete one orbit around the Sun.

Rocky Planets and Gas Giants

The four inner planets — Mercury, Venus, Earth, and Mars — are called the rocky planets, or terrestrial planets. They are small, dense, and have solid, rocky surfaces you could stand on. They formed close to the Sun, where it was too hot for ices and gases to stick together, so only rock and metal could build them. The four outer planets — Jupiter, Saturn, Uranus, and Neptune — are called the gas giants. They are much larger than the rocky planets and are made mostly of hydrogen and helium gas. They formed far from the Sun, where it was cold enough for ice and gas to gather into enormous planets. Jupiter is the largest planet of all — more than 1,300 Earths could fit inside it. Saturn is famous for its beautiful rings, which are made of billions of pieces of ice and rock.

Asteroids, Comets, and Dwarf Planets

Between Mars and Jupiter lies the asteroid belt, a region full of rocky objects called asteroids that never formed into a planet. Comets are chunks of ice, dust, and rock left over from when the solar system formed about 4.6 billion years ago. When a comet travels near the Sun, some of its ice turns to gas and

forms a glowing tail. Scientists study comets because they are like frozen time capsules from the early solar system. Pluto, once counted as the ninth planet, is now classified as a dwarf planet in the Kuiper Belt beyond Neptune.

How Scientists Explore the Solar System

Scientists use telescopes to observe planets, moons, and comets from Earth. Star maps help astronomers chart where objects appear in the night sky throughout the year. Robotic spacecraft and rovers travel to other worlds to collect samples and send back pictures. Everything we know about the solar system comes from careful observation and evidence gathered over many years.

Review Questions

1. What force keeps the planets in orbit around the Sun? 2. List the eight planets in order from closest to the Sun to farthest. 3. Why are the inner planets rocky while the outer planets are gas giants? 4. What is the asteroid belt and where is it located? 5. Why do scientists study comets?